

Policy Gradients in the Complex Plane

*A Natural Gradient Algorithm for Complex-Valued
Reinforcement Learning under Humble Systems Theory*

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Abstract

The companion paper *The Two-Dimensional Agent* (Elwardi, 2026a) establishes a Q-learning framework for complex-valued MDPs (cMDPs) in which the utility $z = c + id$ encodes real cost c and epistemic debt d as the components of a complex number, and the agent minimises $|z|$ without any exploration schedule. That paper is restricted to value-based methods and identifies the contraction of the Bellman optimality operator as its central open problem (OP1).

The present paper extends the framework to *parameterised policies*. We prove a Complex Policy Gradient Theorem (CPGT): the gradient of the complex expected return with respect to policy parameters is given by a REINFORCE-style identity in which the scalar return is replaced by a complex return. A corollary gives a closed-form gradient for the squared modulus objective $|J^\pi(s_0)|^2$, which is directly usable for stochastic gradient descent. We derive the associated natural gradient using a Hermitian Fisher information matrix and propose the **Complex Natural Actor-Critic** (CNAC) algorithm, in which the complex advantage automatically balances cost reduction and epistemic debt reduction without a separate exploration bonus.

We then establish a connection between the gradient flow on $|J^\pi|^2$ and harmonic analysis: the flow lines are steepest-descent curves on a Hermitian manifold. A conjecture is stated—that the Bellman optimality operator restricted to holomorphic value functions contracts in the Bergman metric—which, if proved, would resolve OP1 from the companion paper.

All proven results are labelled **Proven**. Conjectural or heuristic statements are labelled **Conjecture** or **Heuristic** throughout.

“When two objectives are orthogonal, encode them as orthogonal dimensions.”

The Two-Dimensional Agent, Elwardi (2026a)

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1 Introduction

Humble Systems Theory (HST) proposes that every Information Processing System (IPS) evolves toward *epistemic equilibrium*: a state of minimum computational cost subject to informational constraints. The mathematical realisation of this principle in reinforcement learning is the complex MDP (cMDP), in which the immediate utility is a complex number $z = c + id$ encoding real cost c and epistemic debt d , and the agent minimises the modulus $|z|$ rather than maximising a scalar reward.

The companion paper *The Two-Dimensional Agent* [1] develops this framework for *value-based* agents: it proves that the Bellman evaluation operator T^π is a λ -contraction, derives a closed form for the imaginary Q-value as discounted cumulative mutual information, and establishes that $|Q|^2$ is a Lyapunov function under the HST Equilibrium Axiom. The contraction of the Bellman *optimality* operator T remains Open Problem 1 (OP1) of that paper.

The present paper asks a different question: what happens when the agent maintains a *parameterised policy* π_θ rather than a Q-function, and updates θ by gradient descent? This is the policy gradient setting, and extending it to complex returns is not immediate. The standard policy gradient theorem [3] differentiates a scalar expected return; here the return is complex. The objective is the modulus of a complex number, which is not holomorphic, so naïve complex differentiation does not apply.

We resolve this using two tools:

- The **log-derivative trick**, which is measure-theoretic and applies regardless of whether the return is real or complex.
- **Wirtinger calculus** [10], which provides a coherent differential calculus for functions of a complex variable that are not necessarily holomorphic.

The result is a policy gradient identity (Theorem 3.1) and an actor-critic algorithm (Algorithm 1) in which the complex advantage signal automatically drives exploration without any external schedule.

Relationship to OP1. The final section of this paper (Section 7) poses a conjecture that the Bellman optimality operator, when restricted to the space of holomorphic value functions on continuous state-action spaces, is a contraction in the Bergman metric. This is not proved here. It is presented as a research direction that may resolve OP1 by importing tools from complex analysis.

Paper structure. Section 2 recalls the cMDP setup and states conventions. Section 3 proves the Complex Policy Gradient Theorem and its corollaries. Section 4 develops the natural gradient and the Hermitian Fisher information. Section 5 presents the CNAC algorithm. Section 6 establishes the harmonic map connection. Section 7 states the Bergman conjecture. Section 8 collects open problems. Section 9 concludes.

2 Preliminaries

We recall the definitions of [1] that are needed here.

Definition 2.1 (Complex MDP). A cMDP is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, z, \lambda)$ where $\mathcal{S}$ is a (possibly continuous) state space, $\mathcal{A}$ is an action space, $p(s' | s, a)$ is a transition kernel, $\lambda \in [0, 1]$ is a discount factor, and the immediate utility decomposes as:

$$z(s, a, s') = c(s, a, s') + i \cdot d(s, a, s')$$

with $c \geq 0$ a real cost and $d \in \mathbb{R}$ a real epistemic debt.

Definition 2.2 (Complex return and performance). Under a stochastic policy $\pi_\theta(\cdot | s)$ with parameters $\theta \in \mathbb{R}^n$, the complex return from time t is:

$$G_t = \sum_{k=0}^{\infty} \lambda^k \cdot z(S_{t+k}, A_{t+k}, S_{t+k+1}), \quad A_{t+k} \sim \pi_\theta(\cdot | S_{t+k}).$$

The complex expected return from initial state s_0 is:

$$\eta(\theta) := J^\pi(s_0) = \mathbb{E}^\pi[G_0 | S_0 = s_0] \in \mathbb{C}.$$

The **performance** to be minimised is:

$$J(\theta) = |\eta(\theta)| = |J^\pi(s_0)|.$$

Since $|\eta|^2 = \eta\bar{\eta}$ is smooth and minimising $|\eta|$ is equivalent to minimising $|\eta|^2$ (both are zero at the same points and have the same sign of directional derivatives), we work with the squared modulus throughout. This avoids the non-differentiability of $|\eta|$ at zero.

Definition 2.3 (Complex state-value and action-value functions). Under policy π_θ , the complex state-value and action-value functions are:

$$\begin{aligned} V^\pi(s) &= \mathbb{E}^\pi[G_t | S_t = s], \\ Q^\pi(s, a) &= \mathbb{E}^\pi[G_t | S_t = s, A_t = a]. \end{aligned}$$

Both are elements of the Banach space $\mathcal{Q} = \mathcal{B}(\mathcal{S} \times \mathcal{A}, \mathbb{C})$ with sup-norm $\|Q\|_\infty = \sup_{s,a} |Q(s, a)|$. By Theorem 3.1 of [1], Q^π is the unique fixed point of the evaluation operator T^π , which is a λ -contraction.

Definition 2.4 (Complex advantage function). The **complex advantage** at (s, a) under policy π_θ is:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s) \in \mathbb{C}.$$

The complex advantage has both a real component (excess cost relative to the average) and an imaginary component (excess epistemic debt relative to the average). Both enter the gradient update naturally, as shown in Section 5.

3 The Complex Policy Gradient Theorem

3.1 The Log-Derivative Identity

The key insight is that the log-derivative (REINFORCE) identity is measure-theoretic: it requires only that the measure $\mathbb{P}_\theta$ is differentiable in θ and that the integrand has bounded

expectation. The return G_t enters as a complex-valued integrand; the measure remains a real probability measure. This separation is what makes the theorem work.

Let $\tau = (S_0, A_0, S_1, A_1, \dots)$ denote a trajectory under π_θ from $S_0 = s_0$. The probability of a length- T prefix is:

$$\mathbb{P}_\theta(\tau_{0:T}) = \prod_{t=0}^{T-1} p(S_{t+1} \mid S_t, A_t) \cdot \pi_\theta(A_t \mid S_t).$$

The log-derivative is:

$$\nabla_\theta \log \mathbb{P}_\theta(\tau_{0:T}) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t),$$

since the transition kernel p does not depend on θ .

Theorem 3.1 (Complex Policy Gradient Theorem). *(Proven.) For a cMDP with complex utility $z = c + id$, under a stochastic policy π_θ satisfying the regularity conditions of Assumption 3.2, the gradient of the complex expected return is:*

$$\nabla_\theta \eta(\theta) = \nabla_\theta J^\pi(s_0) = \mathbb{E}^\pi \left[\sum_{t=0}^{\infty} \lambda^t G_t \cdot \nabla_\theta \log \pi_\theta(A_t \mid S_t) \mid S_0 = s_0 \right] \in \mathbb{C}^n.$$

Here $G_t = \sum_{k=t}^{\infty} \lambda^{k-t} z(S_k, A_k, S_{k+1})$ is the complex return from time t onward.

Assumption 3.2 (Regularity). 1. For all (s, a) , the map $\theta \mapsto \pi_\theta(a \mid s)$ is differentiable with gradient bounded uniformly: $\|\nabla_\theta \log \pi_\theta(a \mid s)\| \leq B$ for some $B < \infty$.
 2. The immediate utility satisfies $|z(s, a, s')| \leq Z_{\max} < \infty$ for all (s, a, s') , so $|G_t| \leq Z_{\max}/(1 - \lambda)$ almost surely.
 3. The exchange of derivative and expectation is justified (e.g. by dominated convergence).

Proof. Write the complex expected return as an integral over trajectories:

$$\eta(\theta) = \int \left(\sum_{t=0}^{\infty} \lambda^t z_t \right) d\mathbb{P}_\theta(\tau),$$

where $z_t = z(S_t, A_t, S_{t+1})$ and the sum is absolutely convergent by Assumption 3.2(2). Differentiating under the integral:

$$\nabla_\theta \eta(\theta) = \int \left(\sum_{t=0}^{\infty} \lambda^t z_t \right) \nabla_\theta \log d\mathbb{P}_\theta(\tau) d\mathbb{P}_\theta(\tau).$$

The exchange of derivative and integral is justified by Assumption 3.2(3). Applying the log-derivative identity:

$$\nabla_\theta \log d\mathbb{P}_\theta(\tau) = \sum_{t=0}^{\infty} \nabla_\theta \log \pi_\theta(A_t \mid S_t).$$

Rearranging the double sum using the causality of the MDP (the score function at time t is uncorrelated with returns before time t under $\mathbb{P}_\theta$):

$$\begin{aligned}\nabla_\theta \eta(\theta) &= \mathbb{E}^\pi \left[\left(\sum_{t=0}^{\infty} \lambda^t z_t \right) \left(\sum_{s=0}^{\infty} \nabla_\theta \log \pi_\theta(A_s | S_s) \right) \right] \\ &= \sum_{t=0}^{\infty} \mathbb{E}^\pi [G_t \cdot \lambda^t \nabla_\theta \log \pi_\theta(A_t | S_t)],\end{aligned}$$

where the cross terms with $s > t$ vanish by the Markov property (future score functions are independent of past returns given the state), and terms with $s < t$ vanish because $\mathbb{E}^\pi[\nabla_\theta \log \pi_\theta(A_s | S_s) | S_s] = 0$ by the score function identity. This gives the stated result. $\square$

Remark 3.3. The proof is formally identical to the standard (real-valued) policy gradient proof. The only change is that $G_t \in \mathbb{C}$ rather than $G_t \in \mathbb{R}$. The identity holds because: (i) the probability measure $\mathbb{P}_\theta$ is real, (ii) the complex return G_t is integrable by Assumption 3.2(2), and (iii) the score function cancellations rely only on the Markov property and the score identity, both of which are properties of the measure, not of the integrand.

3.2 Gradient of the Squared Modulus

Corollary 3.4 (Gradient of the performance objective). (**Proven.**) *The gradient of the squared modulus objective $J(\theta)^2 = |\eta(\theta)|^2$ with respect to $\theta \in \mathbb{R}^n$ is:*

$$\nabla_\theta |\eta|^2 = 2 \operatorname{Re}(\overline{\eta(\theta)} \cdot \nabla_\theta \eta(\theta)) \in \mathbb{R}^n.$$

Substituting Theorem 3.1:

$$\nabla_\theta |\eta|^2 = 2 \operatorname{Re} \left(\overline{J^\pi(s_0)} \cdot \mathbb{E}^\pi \left[\sum_{t=0}^{\infty} \lambda^t G_t \nabla_\theta \log \pi_\theta(A_t | S_t) \right] \right).$$

Proof. By the product rule for real derivatives:

$$\nabla_\theta |\eta|^2 = \nabla_\theta (\eta \bar{\eta}) = \bar{\eta} \nabla_\theta \eta + \eta \nabla_\theta \bar{\eta} = \bar{\eta} \nabla_\theta \eta + \overline{\bar{\eta} \nabla_\theta \eta} = 2 \operatorname{Re}(\bar{\eta} \nabla_\theta \eta),$$

where the last step uses $w + \bar{w} = 2 \operatorname{Re}(w)$ for any $w \in \mathbb{C}$. Substituting Theorem 3.1 gives the stated form. $\square$

Remark 3.5 (Real output from complex gradient). The gradient $\nabla_\theta |\eta|^2$ is a real vector in $\mathbb{R}^n$, as required for a descent step on $\theta \in \mathbb{R}^n$. The complex structure enters through the inner product $\bar{\eta} \cdot \nabla_\theta \eta$, which weights the policy gradient by the conjugate of the current performance. When J^π is large (the agent is far from equilibrium), the gradient is amplified; as the agent approaches equilibrium and $J^\pi \rightarrow 0$, the gradient magnitude decreases naturally. This is an automatic step-size adaptation arising from the geometry of $\mathbb{C}$.

3.3 Baseline and Variance Reduction

Proposition 3.6 (Complex baseline). *(Proven.) For any $\mathbb{C}$ -valued function $b(s)$, the following identity holds:*

$$\mathbb{E}^\pi[b(S_t) \cdot \nabla_\theta \log \pi_\theta(A_t | S_t)] = 0.$$

Therefore, $\nabla_\theta \eta(\theta)$ is unchanged if G_t is replaced by the complex advantage $A^\pi(S_t, A_t) = G_t - V^\pi(S_t)$:

$$\nabla_\theta \eta(\theta) = \mathbb{E}^\pi \left[\sum_{t=0}^{\infty} \lambda^t A^\pi(S_t, A_t) \cdot \nabla_\theta \log \pi_\theta(A_t | S_t) \right].$$

Proof. By the score function identity, for any fixed s :

$$\mathbb{E}_{A \sim \pi_\theta(\cdot | s)}[\nabla_\theta \log \pi_\theta(A | s)] = \int \nabla_\theta \pi_\theta(a | s) da = \nabla_\theta \int \pi_\theta(a | s) da = \nabla_\theta 1 = 0.$$

Multiplying by $b(s)$ (which is S_t -measurable and thus not affected by the integration over A_t) and taking the full expectation gives the result. $\square$

The complex advantage $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s) \in \mathbb{C}$ has a natural interpretation under HST:

- $\text{Re}(A^\pi(s, a))$ measures the *excess real cost* of action a at state s relative to the average cost of the current policy.
- $\text{Im}(A^\pi(s, a))$ measures the *excess epistemic debt* of action a at state s relative to the average debt of the current policy.

An action with large negative $\text{Im}(A^\pi)$ is one that reduces epistemic debt more than average: it is, by definition, an *informative* action. The gradient update therefore automatically rewards informative actions without any explicit exploration bonus.

4 Natural Gradient in the Complex Domain

4.1 Fisher Information in the Complex Setting

The Euclidean gradient $\nabla_\theta |\eta|^2$ is not invariant to reparameterisation of the policy. The natural gradient [5] corrects for this by pre-multiplying by the inverse Fisher information matrix. In the complex setting, the relevant Fisher information is Hermitian.

Definition 4.1 (Hermitian Fisher information matrix). The **Hermitian Fisher information matrix** of the policy family $\{\pi_\theta : \theta \in \mathbb{R}^n\}$ is:

$$F(\theta) = \mathbb{E}^\pi \left[\nabla_\theta \log \pi_\theta(A_t | S_t) \cdot \overline{\nabla_\theta \log \pi_\theta(A_t | S_t)}^\top \right] \in \mathbb{C}^{n \times n}.$$

Proposition 4.2 (Properties of $F(\theta)$). *(Proven.) $F(\theta)$ is Hermitian ($F = F^\dagger$) and positive semi-definite. If the policy family is identifiable, $F(\theta)$ is positive definite.*

Proof. Hermiticity follows from the definition: $(F)_{jk} = \mathbb{E}[(\nabla_\theta \log \pi_\theta)_j \overline{(\nabla_\theta \log \pi_\theta)_k}]$ and $(F)_{kj} = \mathbb{E}[(\nabla_\theta \log \pi_\theta)_k \overline{(\nabla_\theta \log \pi_\theta)_j}] = \overline{(F)_{jk}}$. Positive semi-definiteness: for any $v \in \mathbb{C}^n$, $v^\dagger F v = \mathbb{E}[|v^\dagger \nabla_\theta \log \pi_\theta|^2] \geq 0$. Strict positivity follows from identifiability. $\square$

Remark 4.3. For real-valued policy families (e.g. softmax over a real parameter), the imaginary parts of $F(\theta)$ are zero and the Hermitian Fisher reduces to the standard real Fisher information matrix. The Hermitian generalisation is non-trivial for complex-parameterised policies such as the ComplexIPS networks of [2], which use Wirtinger backpropagation.

4.2 The Natural Gradient Update

Definition 4.4 (Complex natural gradient). The **complex natural gradient** of $|\eta|^2$ is:

$$\tilde{\nabla}_{\theta}|\eta|^2 = F(\theta)^{-1} \cdot \nabla_{\theta}|\eta|^2 = 2 F(\theta)^{-1} \text{Re} \left(\overline{\eta(\theta)} \cdot \mathbb{E}^{\pi} \left[\sum_t \lambda^t G_t \nabla_{\theta} \log \pi_{\theta}(A_t | S_t) \right] \right).$$

The natural gradient has two advantages over the Euclidean gradient in this setting. First, it is invariant to smooth reparameterisation of θ , so the convergence rate does not depend on the choice of policy coordinates. Second, when cost and debt have very different scales (e.g. $c \sim O(1)$ but $d \sim O(10^3)$ near the start of learning), the Euclidean gradient would be dominated by the imaginary component; the natural gradient compensates through the curvature encoded in $F(\theta)$.

5 The Complex Natural Actor-Critic

We now assemble the theoretical results into a concrete algorithm.

5.1 Architecture

The Complex Natural Actor-Critic (CNAC) consists of:

Critic A function approximator $Q_{\psi} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{C}$ trained to approximate the complex action-value function $Q^{\pi}(s, a)$ under the current policy π_{θ} . Since T^{π} is a λ -contraction (Theorem 3.1 of [1]), the critic can be trained by complex-valued temporal difference learning with convergence guarantees.

Actor A parameterised stochastic policy π_{θ} updated by the natural gradient of $|\eta|^2$ using the complex advantage from the critic.

Fisher estimator An online estimator of $F(\theta)$ using the outer-product approximation, with regularisation $F(\theta) + \epsilon I$ to handle near-singularity.

5.2 Algorithm

5.3 Properties of CNAC

Proposition 5.1 (Automatic exploration-exploitation balance). (*Proven.*) *Under Assumption 6.2 of [1] (the epistemic potential ϕ is a submartingale), the imaginary component of the complex advantage satisfies $\text{Im}(A^{\pi}(s, a)) \in \mathbb{R}$ and encodes the deviation of action a 's epistemic debt from the policy average. No external exploration bonus or schedule is required: actions that reduce epistemic debt below average have negative $\text{Im}(A^{\pi})$, which contributes negatively to $\text{Im}(\nabla_{\theta}\eta)$ and thus drives the actor toward those actions.*

Algorithm 1 Complex Natural Actor-Critic (CNAC)

- 1: **Initialise** actor parameters θ , critic parameters ψ , Fisher estimate $\hat{F} \leftarrow I_n$, step sizes $\alpha_\theta, \alpha_\psi > 0$, regulariser $\epsilon > 0$
- 2: **for** each episode **do**
- 3: Sample trajectory $\tau = (s_0, a_0, z_0, s_1, a_1, z_1, \dots)$ under π_θ
- 4: Compute complex returns: $G_t \leftarrow \sum_{k=t}^{\infty} \lambda^{k-t} z_k$ for each t
- 5: Compute complex TD targets: $Y_t \leftarrow z_t + \lambda Q_\psi(s_{t+1}, \pi_\theta(s_{t+1}))$
- 6: **Critic update** (complex TD):

$$\psi \leftarrow \psi - \alpha_\psi \sum_t \nabla_\psi |Q_\psi(s_t, a_t) - Y_t|^2$$

- 7: Compute complex advantages: $\hat{A}_t \leftarrow G_t - Q_\psi(s_t, a_t)$ (*or* use bootstrapped estimate)
- 8: Compute policy gradient (Proposition 3.6):

$$g \leftarrow 2 \operatorname{Re} \left(\bar{\hat{\eta}} \cdot \sum_t \lambda^t \hat{A}_t \cdot \nabla_\theta \log \pi_\theta(a_t | s_t) \right)$$

where $\hat{\eta} = \sum_t \lambda^t z_t$ is the episode return estimate

- 9: Update Fisher estimate (online):

$$\hat{F} \leftarrow (1 - \beta) \hat{F} + \beta \sum_t \nabla_\theta \log \pi_\theta(a_t | s_t) \overline{\nabla_\theta \log \pi_\theta(a_t | s_t)}^\top$$

- 10: **Actor update** (natural gradient descent):

$$\theta \leftarrow \theta - \alpha_\theta (\hat{F} + \epsilon I)^{-1} g$$

- 11: **end for**
-

Proof. By Definition 2.4 and the structural decomposition of Theorem 3.1 of [1]:

$$\text{Im}(A^\pi(s, a)) = Q_I^\pi(s, a) - V_I^\pi(s),$$

which is the deviation of the action’s discounted cumulative mutual information (Corollary 8.3 of [1]) from the state average. This is non-zero whenever the action is more or less informative than average, and enters the gradient update through $\text{Re}(\bar{\eta} \cdot \nabla_\theta \eta)$ without any additional terms. $\square$

Proposition 5.2 (Epistemic equilibrium as a fixed point). *(Proven, modulo OP1.) At epistemic equilibrium (Axiom 9.1 of [1]), $Q_I^\pi \rightarrow 0$ and therefore $\text{Im}(A^\pi(s, a)) \rightarrow 0$ for all (s, a) . At this point, the CNAC actor gradient reduces to the standard real-valued natural policy gradient applied to the cost component Q_R^π alone. The transition from complex to real gradient is automatic and requires no intervention.*

Remark 5.3 (Comparison with classical exploration). Classical methods such as ε -greedy or entropy regularisation add an explicit exploration term to the objective. This introduces a free parameter (ε , or the entropy weight β) that must be annealed manually and creates a “hockey-stick” training curve: the agent is forced to explore even after it has learned enough. CNAC has no such parameter. The imaginary component of the complex advantage is the exploration signal, and it decays to zero automatically as the critic’s imaginary Q-values approach zero at equilibrium.

6 Harmonic Structure of the Gradient Flow

6.1 The Value Potential Map

Consider the map:

$$\Phi : \mathbb{R}^n \rightarrow \mathbb{C}, \quad \Phi(\theta) = \eta(\theta) = J^\pi(s_0).$$

The gradient flow of CNAC on the squared modulus objective is:

$$\dot{\theta} = -\nabla_\theta |\Phi(\theta)|^2 = -2 \text{Re} \left(\overline{\Phi(\theta)} \nabla_\theta \Phi(\theta) \right).$$

Proposition 6.1 (Descent is orthogonal to phase contours). *(Proven.) The gradient $\nabla_\theta |\Phi|^2$ at any θ points in the direction that most rapidly decreases $|\Phi|$. In the image space $\mathbb{C}$, the descent trajectory is along the radial direction (toward the origin) relative to the current value $\Phi(\theta)$.*

Proof. Write $\Phi(\theta) = |\Phi|e^{i\alpha}$ for phase $\alpha = \text{Arg}(\Phi)$. Then $|\Phi|^2 = |\Phi|^2$ depends only on the modulus. The directional derivative of $|\Phi|^2$ in direction v is $2 \text{Re}(\overline{\Phi} \langle \nabla_\theta \Phi, v \rangle)$, which is maximised when $\langle \nabla_\theta \Phi, v \rangle$ is parallel to Φ in $\mathbb{C}$, i.e. when the image-space increment points radially. $\square$

6.2 Connection to Harmonic Analysis

The following result characterises when the objective $U(\theta) = |\Phi(\theta)|^2$ is harmonic (i.e. $\Delta_\theta U = 0$).

Proposition 6.2 (Laplacian of the objective). (*Proven.*) The Laplacian of $U(\theta) = |\Phi(\theta)|^2$ with respect to $\theta \in \mathbb{R}^n$ is:

$$\Delta_\theta U = 2 \|\nabla_\theta \Phi\|^2 + 2 \operatorname{Re}(\bar{\Phi} \Delta_\theta \Phi),$$

where $\|\nabla_\theta \Phi\|^2 = \sum_{j=1}^n |\partial_{\theta_j} \Phi|^2$.

Proof. By the product rule: $\partial_j U = \partial_j(\Phi \bar{\Phi}) = (\partial_j \Phi) \bar{\Phi} + \Phi \overline{(\partial_j \Phi)} = 2 \operatorname{Re}(\bar{\Phi} \partial_j \Phi)$. Differentiating again: $\partial_j^2 U = 2 |\partial_j \Phi|^2 + 2 \operatorname{Re}(\bar{\Phi} \partial_j^2 \Phi)$. Summing over j gives the result. $\square$

Proposition 6.3 (Harmonicity condition). (*Proven.*) $U(\theta) = |\Phi(\theta)|^2$ is harmonic if and only if:

$$\operatorname{Re}(\bar{\Phi}(\theta) \Delta_\theta \Phi(\theta)) = - \|\nabla_\theta \Phi(\theta)\|^2.$$

In particular, U is superharmonic (satisfies $\Delta U \leq 0$) if:

$$\operatorname{Re}(\bar{\Phi} \Delta_\theta \Phi) \leq - \|\nabla_\theta \Phi\|^2.$$

Remark 6.4 (HST interpretation). Superharmonicity of $U = |\Phi|^2$ is a strong condition: it implies that U cannot have interior local maxima, and that the gradient flow is a descent toward a global minimum. Whether the value potential map $\Phi(\theta) = J^\pi(s_0)$ satisfies the superharmonicity condition depends on the structure of the policy family and the cMDP. We conjecture (but do not prove) that it holds for well-conditioned policy families at epistemic equilibrium, where $\operatorname{Im}(\Phi) \rightarrow 0$ and the condition reduces to a statement about the real cost function.

6.3 Kähler Geometry: A Heuristic Connection

Status: Heuristic. The following is not proved. It is presented as a geometric analogy that motivates the Bergman conjecture of Section 7.

If we formally *complexify* the parameter space by extending $\theta \in \mathbb{R}^n$ to $\zeta \in \mathbb{C}^n$ (treating the real parameters as the real parts of complex coordinates), then the map $\zeta \mapsto \Phi(\zeta)$ becomes a map from a complex manifold to $\mathbb{C}$. The Hermitian Fisher information matrix $F(\theta)$ can be viewed as a Hermitian metric on this complexified parameter space. A Hermitian metric h on a complex manifold is Kähler if its associated 2-form ω_h is closed: $d\omega_h = 0$.

Heuristic. If $F(\theta)$ arises as the Hessian of a Kähler potential $K(\zeta, \bar{\zeta})$ on the complexified parameter space, then the natural gradient flow of CNAC is a gradient flow on a Kähler manifold, and the following properties hold:

- The flow is volume-preserving on level sets of $\operatorname{Im}(\Phi)$ (since Kähler flows are Hamiltonian with respect to the symplectic form ω_h).
- The cost-debt coupling is encoded in the Kähler potential K , which plays the role of the HST energy budget constraint.
- Epistemic equilibrium corresponds to a zero of the moment map associated with the symplectic action of the phase rotation group $U(1) \curvearrowright \mathbb{C}$.

These properties are geometrically natural and consistent with the HST framework. Whether $F(\theta)$ is in fact the Hessian of a Kähler potential depends on the policy family and is an open question.

7 The Bergman Conjecture and Open Problem 1

We now state the conjecture that may resolve OP1 of [1].

7.1 Background: The Bergman Metric

Let $\Omega \subset \mathbb{C}$ be a bounded domain. The **Bergman space** $\mathcal{A}^2(\Omega)$ consists of holomorphic functions $f : \Omega \rightarrow \mathbb{C}$ that are square-integrable with respect to the Lebesgue measure on Ω :

$$\mathcal{A}^2(\Omega) = \left\{ f : \Omega \rightarrow \mathbb{C} \text{ holomorphic} \mid \int_{\Omega} |f(z)|^2 dA(z) < \infty \right\}.$$

The **Bergman metric** on Ω is the Hermitian metric induced by the Bergman kernel $K(z, w)$ of $\mathcal{A}^2(\Omega)$. Crucially, the Bergman metric is complete and its curvature is well-controlled on bounded domains.

7.2 The Conjecture

Recall that OP1 of [1] asks whether the Bellman optimality operator:

$$(TQ)(s, a) = \sum_{s'} p(s' \mid s, a) [z(s, a, s') + \lambda Q(s', \pi_Q(s'))]$$

is a contraction on $(\mathcal{Q}, \|\cdot\|_{\infty})$. The geometric obstruction identified in [1] is the absence of a total order on $\mathbb{C}$ and the phase cancellation between nearly-optimal complex values.

The Bergman conjecture proposes a resolution by changing the space.

Open Problem 1 (Bergman contraction conjecture). Suppose the state-action space $\mathcal{S} \times \mathcal{A}$ is a bounded domain $\Omega \subset \mathbb{C}$ (or a product of such domains). Restrict the Bellman optimality operator T to the Bergman space:

$$T_{\mathcal{A}} : \mathcal{A}^2(\Omega) \rightarrow \mathcal{A}^2(\Omega), \quad (T_{\mathcal{A}}Q)(s, a) = (TQ)(s, a).$$

Conjecture: $T_{\mathcal{A}}$ is a contraction in the Bergman norm $\|Q\|_{\mathcal{A}} = \left(\int_{\Omega} |Q(z)|^2 dA(z) \right)^{1/2}$.

Motivation. Three properties of the Bergman space support this conjecture:

1. **Holomorphic constraint eliminates phase cancellation.** Holomorphic functions on a connected domain cannot have arbitrary phase behaviour: the argument principle and the open mapping theorem constrain the range. In particular, the modulus-greedy selector $\arg \min_a |Q(s, a)|$ may become continuous (in the Bergman topology) when Q is holomorphic, removing Face 2 of the obstruction identified in [1].
2. **Compactness.** The inclusion $\mathcal{A}^2(\Omega) \hookrightarrow C(\Omega)$ is compact on bounded domains (Montel's theorem for square-integrable functions), so $T_{\mathcal{A}}$ maps bounded sets to relatively compact sets. Schauder's fixed-point theorem then gives existence of a fixed point without requiring a contraction.
3. **Curvature control.** The Bergman metric has negative holomorphic sectional curvature on bounded symmetric domains. This is the geometric analogue of the contractivity of the operator: negative curvature means that geodesics converge.

What needs to be proved.

1. That the modulus-greedy selector $\pi_Q(s) = \arg \min_a |Q(s, a)|$ maps holomorphic Q to a selector that is measurable (and ideally continuous) in the Bergman topology.
2. That the composition $Q \mapsto Q \circ \pi_Q$ maps $\mathcal{A}^2(\Omega)$ into itself (i.e. that the composed function is holomorphic).
3. That $\|T_{\mathcal{A}}Q_1 - T_{\mathcal{A}}Q_2\|_{\mathcal{A}} \leq \kappa \|Q_1 - Q_2\|_{\mathcal{A}}$ for some $\kappa < 1$.

Remark 7.1. The Bergman conjecture requires a continuous state-action space, so it does not directly resolve OP1 in the finite-state tabular setting of [1]. However, a positive resolution for continuous domains would provide strong evidence that the contraction gap of OP1 is an artefact of the sup-norm on discrete spaces, not a fundamental obstruction. It would also connect complex RL to the rich theory of holomorphic dynamics and Kobayashi hyperbolicity.

8 Open Problems

Open Problem 2 (Convergence of CNAC). Does the CNAC algorithm (Algorithm 1) converge almost surely to a local minimum of $|\eta(\theta)|^2$? A sufficient condition would be that the complex policy gradient $\nabla_{\theta}\eta$ is Lipschitz in θ , which holds if the policy family is sufficiently regular. The standard two-timescale ODE analysis of actor-critic methods [9] requires the critic to converge faster than the actor; in the complex setting, the critic convergence rate is governed by the contraction constant of T^{π} , which is λ (Theorem 3.1 of [1]).

Open Problem 3 (Sample complexity of complex advantage estimation). The complex advantage $A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$ is estimated from samples. The real part has classical variance bounds (analogous to standard advantage estimation). The imaginary part—the epistemic debt deviation—has variance that depends on the rate of information gain along trajectories. Characterise the sample complexity of complex advantage estimation as a function of the epistemic potential ϕ and the mixing time of the Markov chain under π_{θ} .

Open Problem 4 (Kähler condition on the Fisher metric). Under what conditions on the policy family π_{θ} is the Hermitian Fisher information matrix $F(\theta)$ the Hessian of a Kähler potential on the complexified parameter space? For exponential families, the Fisher metric is the Hessian of the log-partition function, which is real. Complexification of exponential families (e.g. via complex natural parameters) may yield a Kähler structure; this is related to the theory of information geometry on complex statistical manifolds [6].

Open Problem 5 (Bergman contraction for tabular cMDPs). The Bergman conjecture (Open Problem 1) requires continuous domains. Find an analogue for finite-state cMDPs: identify a norm or weighted inner product on $\mathcal{B}(\mathcal{S} \times \mathcal{A}, \mathbb{C})$ under which the Bellman optimality operator T is a contraction. The phase-weighted norm $\|Q\|_{\infty, \beta} = \max_{s,a} e^{\beta |\text{Arg}(Q(s,a))|} |Q(s, a)|$ is a candidate (cf. OP1(c) of [1]).

9 Conclusion

We have extended the complex-valued reinforcement learning framework of Humble Systems Theory from value-based to policy-based methods. The central result is the Complex Policy Gradient Theorem (Theorem 3.1): the gradient of the complex expected return is given by a REINFORCE-style identity with complex returns, and the gradient of the squared modulus objective is the real part of a Hermitian inner product (Corollary 3.4).

The CNAC algorithm (Algorithm 1) implements natural gradient descent on this objective. Its key property is that the complex advantage signal automatically encodes both cost optimisation and epistemic debt reduction, so no separate exploration bonus or schedule is required. As the critic’s imaginary Q-values decay to zero at epistemic equilibrium, the actor gradient reduces automatically to the standard real-valued natural policy gradient for pure cost minimisation.

The harmonic analysis of Section 6 identifies the gradient flow as steepest descent on a Hermitian manifold, and the Kähler heuristic of Section 6.3 connects the Fisher information to the geometry of the cost-debt plane. These connections are heuristic and not proved; they serve as geometric motivation.

The Bergman conjecture (Open Problem 1) is the principal open problem of this paper. It proposes that restricting the Bellman optimality operator to holomorphic value functions resolves the contraction gap identified in OP1 of the companion paper. The three properties cited—holomorphic constraint on phase, compactness in the Bergman norm, and negative curvature of the Bergman metric—are each necessary for the argument but not jointly sufficient. Closing this gap is the next step.

*The geometry of $\mathbb{C}$ provides the exploration-exploitation trade-off for free.
The natural gradient provides the reparameterisation invariance for free.
The open problems provide the next two years of work.*

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